CataTriage AI Two Page Summary

CataTriage AI is a working research prototype for reviewing cataract related information and coordinating a doctor approved hospital workflow. It combines two real data research models, protected doctor admin and technician portals, four auditable processing stages and a native doctor Android app. Its measured model performance is limited, so it does not make clinical decisions.

What a user can do

User	Implemented functions
Doctor	Review assigned patients and per eye original images, record symptoms and notes, inspect results, enter a tier, approve a month, message, manage profile and password.
Hospital admin	Manage accounts and patients, inspect approved queue and capacity, choose a free date inside the approved month, resolve leave and view audit history.
Technician	Register structured information and upload left right or unknown side images with immediate technical quality feedback.

How the system works

Preprocessing checks the input quality and make sure all needed information is there. Inference runs the model based on the selected modality. Safety checks identify uncertainty and conditions that are not supported. Reporting shows the results and audit trail for doctors to review. These are fixed software steps and not separate medical agents.

Scheduling is done by doctor tier and month approval first and then the admin confirms the exact date. Model confidence does not decide the urgency or how long a patient can safely wait. If there is any change in the evidence a new month review is needed. The floating red bell stays visible when months are approved but exact dates are not yet confirmed. A built in navigation guide also provides links based on user roles and helps with the workflow model metrics and image capture checklists.

Stack and delivery

FastAPI and Jinja2 provide one Python web and mobile API codebase. PyTorch and exported NumPy tree inference run on CPU or CUDA. Three SQLite files support local use; cloud mode uses three private Supabase schemas. Vercel hosts the app. Optional Gemma explanations have a configured free fallback. The doctor app is native Java Android 8 or newer, with server side inference.

Source code, model checkpoints, SQL, tests, training utilities, APK and setup instructions accompany this summary. The local demo starts without raw datasets, cloud keys or production passwords after dependencies are installed. Every supplied PDF has an editable Word counterpart.

Implementation Results and Model Evidence

Model	Training	Validation	Test	Previous test
Grading v2	70.79%	54.36%	56.04%	53.69%
Fusion v2	72.14%	65.00%	55.00%	52.50%

Grading gained seven exact predictions on 298 test participants; fusion gained one on 40 images. Both also improved test macro F1. The intervals are wide and overlap prior scores: grading about 50.7 to 61.7 percent, fusion 37.5 to 71.4 percent. Normal recall is still only 2 of 8. A trial with 99 percent training accuracy was rejected for validation overfitting.

The data and its age

Grading uses 1,986 participants from Taizhou study which was done from July to September 2012 and published in Dryad in 2016. The data is divided into 1,390 for training, 298 for validation and 298 for testing. This is only one retrospective cohort with a maximum grade taken from both eyes. It is not validated using multiple clinical cohorts.

Fusion uses 200 clean images from the public CatScreen sample which was downloaded on 26 October 2026. The data is divided into 140 training, 20 validation and 40 testing images with different participant IDs in each set. The collection dates were not confirmed from the sample. It only provides acquisition metadata and not the full clinical history. Both cohorts are not linked with each other and no chronological external or prospective validation has been done.

Honest comparison with the proposal

The interface, human approval workflow, role stores, reports, messaging, profile features, alerts, chatbot and doctor APK are already implemented. But verified image linked severity, full clinical fusion, a trained urgency or safe waiting model and clinical deployment readiness are still not proven. Automatic WhatsApp messaging needs a verified Business configuration. The navigation guide is not a medical LLM. Synthetic workflow fixtures are also different from real model evidence.

How judges can inspect it

Read the implementation report, run the README local setup and tools/judge_smoke.py, and try doctor month approval followed by admin date selection. Inspect the current model evidence and four agent trace, then try “open calendar” in the guide. The recorded workspace run passed 110 backend tests; a source only checkout skips one optional raw image test. Browser checks and live non persistent inference also passed.

- Website: https://catatriage-ai.vercel.app
- Repository : https://github.com/CodeJediX/CataTriage-AI
- Data sources https://doi.org/10.5061/dryad.gt410 and the official CatScreen dataset page.
- The local fixture accounts are in README. Hosted account credentials are not distributed.
- Android Doctor App Download (APK v1.1.0) : https://raw.githubusercontent.com/CodeJediX/CataTriage-AI/main/output/android/CataTriage-Doctor-1.1.0.apk
- Demo Video Link :https://drive.google.com/drive/folders/1W5UWquWD7KfDNJacaq0Vw75JG4xaJx-0?usp=sharing